

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Smart Complaint Management System

Campus Facility Complaint Registration, Routing and Resolution Tracking

SOFTWARE REQUIREMENTS SPECIFICATION

Item	Details
Group / Course	Team 6 / UE24CS341A — Software Engineering
Document version	2.0 — consolidated; product behaviour retained
Product scope	SCMS Release 1.0; six planned Agile sprints
Document status	**Prepared for submission**

Team profile

Member / ownership	SRN / PRN
Anushka K Kishore — F1	PES1UG24CS074 / PES1202400461
Atul Suresh — F2	PES1UG24CS098 / PES1202402181
Chaitanya Krishna Makkena — F3	PES1UG24CS122 / PES1202403729
Tanmay Sridhar — F4	PES1UG25CS849 / PES1202503400

Contents and introduction

Section	Pages
1. Introduction / document control	2
2. Overall Description	3
3. External Interface Requirements	4
4. Analysis Models and consolidated use cases	5–8
5. System Features	9–12
6. Other Nonfunctional Requirements	13–15
7. Other Requirements	15
Appendix A — Glossary	15
Appendix B — Field Layouts	16
Appendix C — Requirement Traceability Matrix	17–18

1.1 Purpose and audience

Specify SCMS Release 1.0 for Team 6, course evaluators and the Campus Facility and Hostel Administration Office. Requirements cover complaint registration, routing, assignment, resolution, tracking and reporting.

1.2 Scope and reading guide

SCMS records, routes, assigns, resolves and tracks campus/hostel complaints, including evidence, anonymous reporting, SLA escalation, notifications, feedback and reporting. Developers use §§4–6 and Appendix C; reviewers start with §§1–3 and the use-case model. Physical repairs and emergency response are outside the application.

Excluded: AI/ML, native apps, SMS/WhatsApp, ERP/HRMS, payments/procurement, IoT, SSO/biometrics, live chat, offline/multilingual operation, multi-campus tenancy, GPS/maps and staff rostering.

1.3 Document control

Version	Description
1.0 — 2 Sep 2026	Original project requirements baseline.
1.1 — 1 Oct 2026	Course CI/CD correction and security/actor clarifications.
2.0 — consolidated	10 use cases; 57 requirement/rule IDs; main-section integration and updated traceability.

2. Overall Description

2.1 Product perspective and functions

SCMS is a standalone campus web portal replacing an informal complaint record process. It does not connect to ERP or perform repair work. Main functions are access/RBAC, authenticated/anonymous intake and evidence, own/reference tracking, routing/queues/allocation, due-date escalation, controlled resolution/feedback/closure, notices and administrator/department reports.

2.2 User classes

Actor	Authority
Complainant	Register/login; submit/track own complaints; confirm/reject own resolution.
Anonymous Complainant	No login/registration; submit, restricted tracking and permitted resolution feedback by valid reference.
Department Staff	Resolver; own-department queue and assigned progress/resolution; state-model actions only.
Department Head	Own-department allocation, reassignment, permitted state actions, escalation and reports. Warden is this role, not a fifth role.
Administrator	Users/roles/membership, master data/routing/SLA, campus reports and state-model rejection.
Scheduler / Mail Service	Secondary systems for escalation/auto-close and outbound email.

2.3 Operating environment

Modern desktop/mobile browsers, current or preceding major Chrome/Firefox/Edge/Safari; minimum 320 px viewport. Linux server; Python 3.x/Django; Docker application; PostgreSQL deployment and SQLite local development. Django file storage, SMTP relay and a scheduler invoking management commands. Campus/internet browser access uses HTTPS.

2.4 Constraints

Use Agile, Git/GitHub and GitHub Projects with genuine member contributions. Keep F1–F4 end-to-end ownership and unit-testable business logic outside templates. Jenkins CI/CD shall build, test and deploy to a test environment; code review, security validation, static analysis and line/branch reporting are required.
Master-data changes require configuration only; no paid service is required.

2.5 Assumptions and dependencies

Single-campus English deployment; network/email/scheduler/database/storage availability. Categories, departments and SLA values are provisional pending stakeholder review. Expected use is a few thousand users/few hundred monthly complaints with low concurrency; this is an assumption, not measured adoption. Django, PostgreSQL, SMTP, scheduler and test/CI tools are dependencies. No uptime percentage or working-hour/holiday calendar is promised.

3. External Interface Requirements

3.1 User interfaces

Interface	Required behaviour
Access and header	Registration/login; role landing page; signed-in name/role, notices/unread count and logout.
Intake and acknowledgement	Category/location/title/description/priority and optional evidence; clear ID; anonymous one-time reference; emergency-channel warning.
Tracking and work	Own complaints/filter/detail/history; reference lookup; department queue; progress/resolution; Head allocation/monitoring.
Administration/reports	Master data/users/roles/rules/SLA; campus or department report and CSV.
Shared UI rules	Responsive >=320 px; text status labels; field errors preserving safe values; copyable ID; at most 25 rows/page with status/category filters; explicit confirmation for irreversible actions.

3.2 Software interfaces

Interface	Direction / data / purpose
Database	Django ORM reads/writes users, complaints, configuration, evidence metadata, history and notifications. PostgreSQL deployed; SQLite local; migration-controlled schema.
File storage	Django storage handles evidence bytes; database holds metadata/reference. Required storage failure rejects intake; protected download only.
SMTP	Outbound recipient/subject/complaint ID/status via Django email backend and encrypted relay. No inbound email processing.
Scheduler	Invokes escalation and resolved-complaint auto-close management commands.
Development tools	Jenkins, pytest/coverage and static analysis read code/test artefacts; not runtime features.

3.3 Communications

Serve HTML/forms over HTTPS and redirect HTTP. POST for state-changing forms with valid CSRF; no state-changing GET. Secure/HttpOnly/SameSite session cookies; multipart evidence uploads enforce configured limits. SMTP uses STARTTLS/SMTPS. Credentials/endpoints come from environment configuration. Email excludes credentials, session tokens and anonymous references. No inbound third-party integration or hardware interface is required.

Public REST APIs and native binary interfaces are not established. SAD documents web form/route contracts and internal Python component interfaces instead.

4. Analysis Models

4.1 Actor terminology and intake view

Actor names in diagrams and text are identical. Department Staff is the system role commonly called resolver. Anonymous Complainant is separate and has no association with registration/login. Public feedback is limited to a valid anonymous complaint reference.

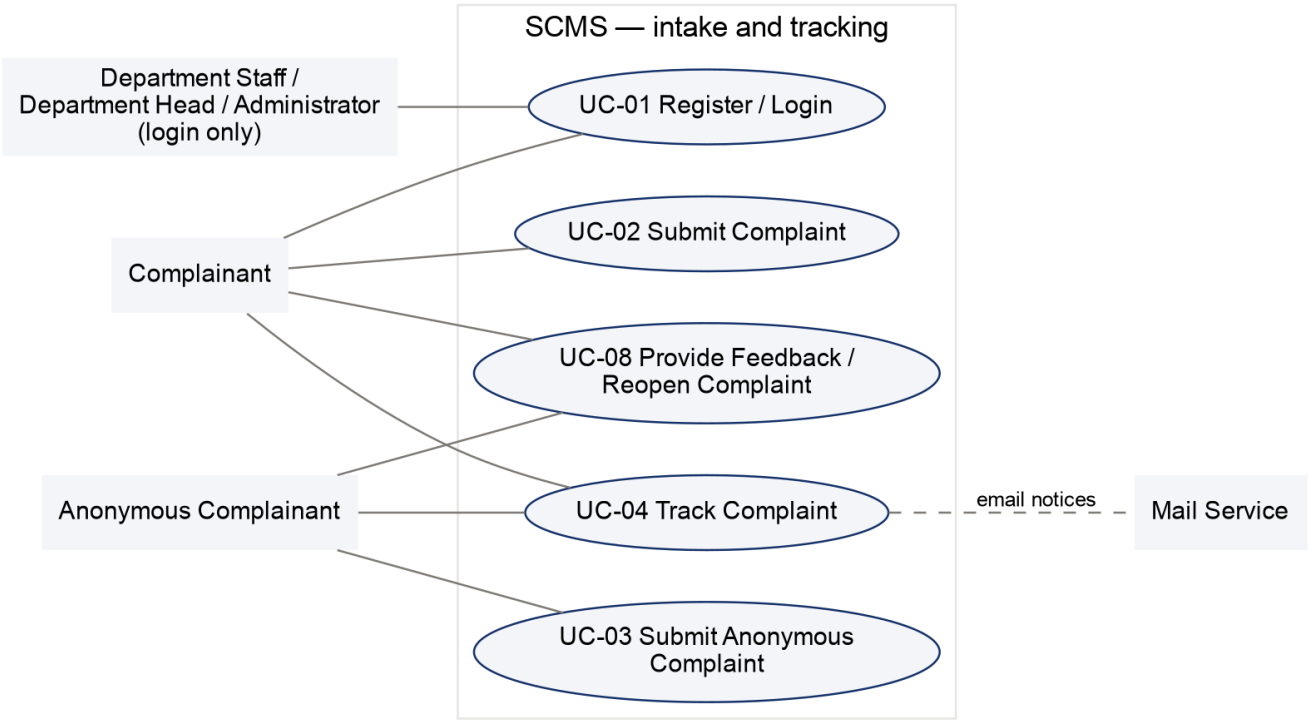


Figure SRS-1. Intake, tracking and feedback use cases. Other role actors associate with UC-01 for login only.

Routing (UC-05) is a system step of UC-02/UC-03, not a scheduler action. Notifications appear within tracking and workflow use cases rather than as a separate button-level use case. The two views together cover UC-01–UC-10.

4.2 Use-case organization

The 10 use cases group related complaint-workflow goals. Evidence is covered by intake and resolution; queues and reassignment by allocation; notifications by tracking and workflow; auto-close by feedback; and master-data actions by configuration. Detailed requirement clauses are specified in §5 and §6.

4. Analysis Models — operations

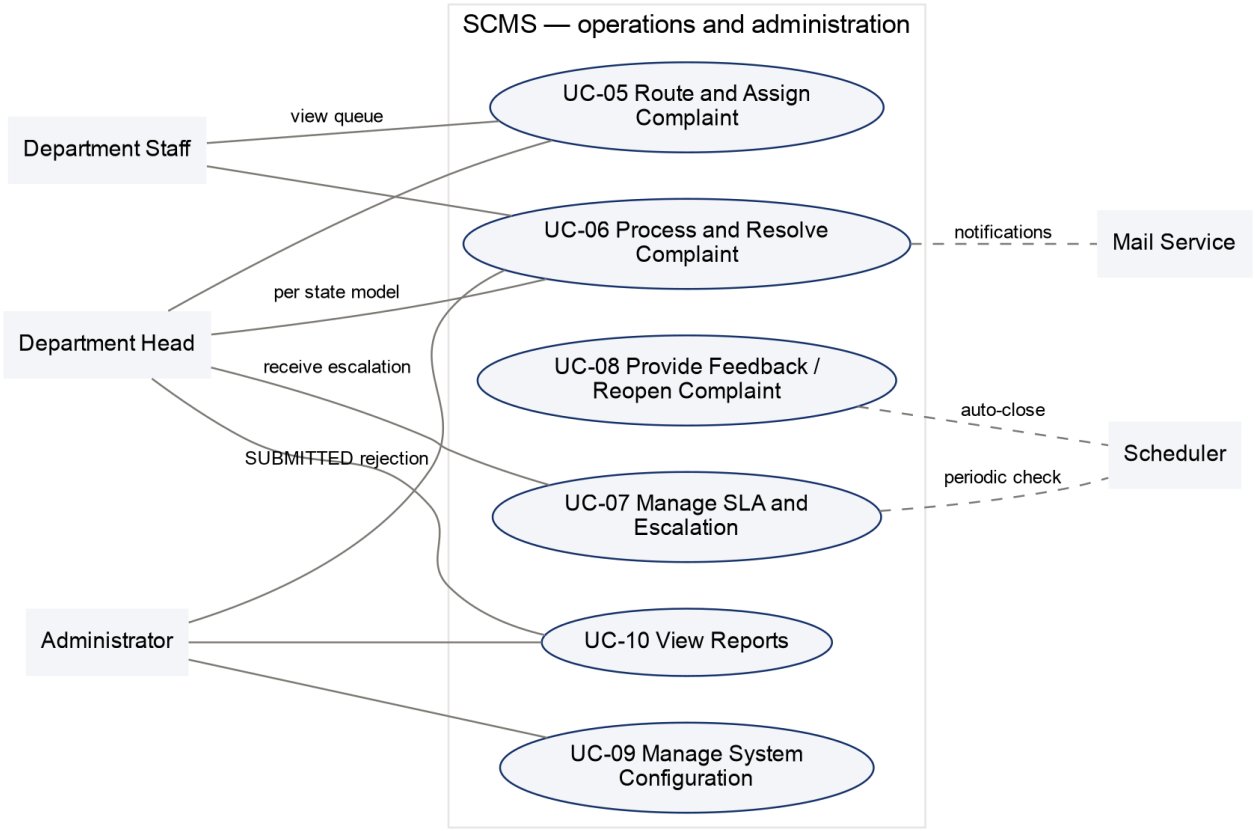


Figure SRS-2. Operations, configuration and reporting; Scheduler handles escalation and auto-close only.

4.3 Complaint state model

From	To	Permitted actor
SUBMITTED	ASSIGNED / REJECTED	System routing / Administrator or Department Head rejection
ASSIGNED	IN_PROGRESS / REJECTED	Assigned Department Staff or Department Head
IN_PROGRESS	RESOLVED	Assigned Department Staff
RESOLVED	CLOSED / REOPENED	Originating Complainant or valid anonymous reference holder; Scheduler closes after expiry
REOPENED	IN_PROGRESS	Assigned Department Staff or Department Head
CLOSED / REJECTED	None — terminal	No further transition

ASSIGNED means department-routed; individual allocation changes the assignee, not state. ESCALATED is an independent flag/level on open states, not a state. Queue visibility never authorizes unassigned Staff to progress/resolve. Head/Administrator actions are only those shown above; no additional resolver permissions are implied.

4.4 Consolidated use cases — intake and work

UC-01 — Register / Login

Actor: Complainant; Department Staff, Department Head and Administrator log in only. Precondition: Application available; account exists for login.

Flow: Complainant registers with validated fields and receives the default role. User logs in to the role page; logout ends session.

Alternative/error: Duplicate email/invalid passwords or credentials are rejected appropriately; lockout and idle expiry apply.

UC-02 — Submit Complaint

Actor: Complainant Precondition: Logged in; active master data/default routing ready.

Flow: Enter category, location, title, description and priority; optionally attach evidence. Validate, route, calculate due time and display unique ID.

Alternative/error: Missing/inactive/invalid fields or files reject intake. Failed storage/routing leaves no accepted record; retry is not a duplicate.

UC-03 — Submit Anonymous Complaint

Actor: Anonymous Complainant Precondition: No registration or login; intake settings available.

Flow: Submit the same complaint form without identity. Apply normal validation/routing and show ID plus a distinct one-time tracking reference.

Alternative/error: Invalid intake is rejected; no identifying account/name/email/IP is recorded. Lost reference is not recoverable.

UC-04 — Track Complaint

Actor: Complainant; Anonymous Complainant Precondition: Owner session, or valid anonymous reference.

Flow: Owner views filtered list, authorized detail/history/evidence and notices. Reference holder sees only permitted status fields/history.

Alternative/error: Other-owner/department records denied. Unknown/malformed references have the same no-match; lookup limit is 10/source/10 minutes.

UC-05 — Route and Assign Complaint

Actor: Department Head; Department Staff (queue); system during intake. Precondition: Complaint accepted; rules/default and department members configured.

Flow: Automatic routing places complaint as ASSIGNED in a department queue. Head assigns/reassigns within department; queue orders escalation, priority, age.

Alternative/error: No match uses flagged default. Inactive rules ignored; other-department assignee denied; reassignment needs reason and retains state/history.

4.4 Consolidated use cases — completion and administration

UC-06 — Process and Resolve Complaint

Actor: Department Staff; Department Head/Administrator only where state model permits. Precondition: Authorized department/assigned work; valid current state.

Flow: Progress ASSIGNED/REOPENED to IN_PROGRESS; record remarks and optional evidence to resolve. Authorized rejection follows the state model; append history.

Alternative/error: Invalid actor/state, short resolution remarks or competing change refused; no history edits/deletion. Resolver cannot close own work.

UC-07 — Manage SLA and Escalation

Actor: Scheduler; Department Head receives escalation. Precondition: Stored priority/category due time; scheduled command available.

Flow: Check eligible overdue complaints; flag/increment level within interval/max and notify Head. Record run outcome.

Alternative/error: Exclude resolved/terminal cases. Repeat/overlapping/interrupted runs do not repeat completed escalation; no automatic reassignment.

UC-08 — Provide Feedback / Reopen Complaint

Actor: Complainant; Anonymous Complainant with valid reference; Scheduler for auto-close. Precondition: Complaint RESOLVED; originator/reference authority.

Flow: Confirm, optionally rate 1–5, then close; or reject with reason and reopen same department. Scheduler closes unconfirmed after default 7 days.

Alternative/error: Reason <20 characters/invalid rating/wrong owner denied. Default maximum 2 reopens; further rejection raised to Head. Never auto-close a REOPENED complaint.

UC-09 — Manage System Configuration

Actor: Administrator Precondition: Administrator session; valid configuration.

Flow: Manage users/roles/membership, categories/locations/departments, routing/SLA configuration; audit changes.

Alternative/error: Protect last Admin, disallow self-role changes, reject duplicate active rules/nonpositive durations, deactivate referenced data rather than delete.

UC-10 — View Reports

Actor: Administrator; Department Head Precondition: Authorized reporting role; selected date range.

Flow: View grouped volumes, open/overdue/escalated/reopened counts, resolution mean/median and authorized reopen information; export matching CSV.

Alternative/error: Head scope limited to own department; no anonymous attribution/reference export. Report formula/date-boundary decisions require confirmation.

5. System Features

5.1 User and Access Management

High priority; F1. Registration/login establishes the current role and scope. Denied requests cannot change protected data.

ID / capability	Requirement
REQ-AUTH-01 Registration	The system shall register a visitor using full name, unique email, password and user type (student/faculty/staff), with hostel block and room for hostel residents. Passwords require at least 8 characters, including a letter and a digit. New accounts receive the Complainant role; users cannot choose or change their own role. Duplicate email shall create no account and receive the same generic acknowledgement as successful registration, directing existing-account users to password-reset assistance (its flow is unspecified).
REQ-AUTH-02 Login and lockout	The system shall authenticate registered email/password, create a session and show the role landing page. Invalid credentials receive a generic failure. Five consecutive failures within 15 minutes shall block authentication for a configurable period, default 15 minutes.
REQ-AUTH-03 Session termination	The system shall invalidate a session on logout or after configurable inactivity, default 30 minutes, and require login for subsequent protected actions.
REQ-ACC-01 Role and complaint scope	Each authenticated user shall have exactly one role: Complainant, Department Staff, Department Head or Administrator. Every request shall check current role and department membership. Complainants read only their own complaints; Department Staff read only their department queue and progress/resolve only assigned work. Department Head powers follow the state model; other-department access is refused.

5.2 Complaint Management

High intake/tracking; Medium anonymous intake. F2 validates records/evidence/references; F4 supplies tracking.

ID / capability	Requirement
REQ-CMP-01 Complaint form	The system shall accept category, optional sub-category, block/floor/room or area, title, description and indicated priority. Category, location, title and description are mandatory; description is at least 20 characters. Only active, known category/location values are accepted. Field limits apply; rejection shall preserve other safe entered values and explain the error.
REQ-CMP-02 Acknowledgement and record	Each accepted complaint shall receive a unique immutable ID SCMS-YYYY-NNNNNN, displayed on acknowledgement. Record submission time and initial SUBMITTED state, and the account for authenticated intake only. Routing then changes the state to ASSIGNED.

5.2 Complaint Management (continued)

ID / capability	Requirement
REQ-CMP-03 Evidence validation	Submission and resolution shall allow up to 3 JPEG, PNG or PDF evidence files, each at most 5 MB. Validate actual content type, not extension/client declaration; store files under generated names. Invalid count, size or type shall reject the operation with a reason.
REQ-CMP-04 No partial or duplicate intake	Complaint creation, evidence storage and initial routing shall succeed together. Failure shall leave no complaint, accepted identifier or attachment. Retrying a single submission after a timeout shall not create a duplicate complaint.
REQ-ANON-01 Anonymous intake and reference	An Anonymous Complainant shall submit without registration/login using the same validation, evidence, routing and duplicate/failure rules as authenticated intake, except no submitting account. Issue an unguessable tracking reference distinct from the complaint ID; display it once with a notice that it cannot be recovered.
REQ-ANON-02 Anonymous privacy	No anonymous complaint shall store account, name, email or client IP, or become attributable through an application view/export/report. Staff views shall identify the reporter as anonymous. Rate-limit/security metadata shall not be stored against or linked to the complaint.
REQ-ANON-03 Restricted anonymous tracking	A valid reference shall reveal only that complaint's ID, category, location, current status, escalation flag and status history. Unknown and malformed references receive the same no-match response. Allow at most 10 tracking attempts per request-level source in 10 minutes; the source key shall not be persisted or linked to the complaint.
REQ-TRK-01 Tracking and detail	The Complainant shall see a status/category-filtered list of own complaints with ID, category, location, status, escalation, submission and due times. Authorized complaint detail shall show submitted data, evidence, state, department, escalation level and chronological history. Anonymous tracking remains restricted by REQ-ANON-03.

5.3 Assignment and Resolution

High priority; F3/F4. Intake triggers department routing and due-date calculation; Head allocation and Staff resolution follow.

ID / capability	Requirement
REQ-ROUTE-01 Automatic routing	The system shall route every accepted complaint using active rules and departments: category+location before category-only before location-only, then highest configured precedence. No match uses the configured default department and an unrouted flag for Administrator attention. Record department, applied rule/default and time; place the complaint in that queue as ASSIGNED. Rule changes affect only new complaints. Department assignment is not individual resolver assignment.

5.3 Assignment and Resolution (continued)

ID / capability	Requirement
REQ-ASSIGN-01 Queue and resolver allocation	Department queues shall order escalated complaints first, then priority, then oldest submission. The Department Head shall assign ASSIGNED work to Department Staff in the same department, or reassign already allocated work within that department with a reason. Record actor, assignee, time and reassignment reason; allocation does not change state. Cross-department targets are refused.
REQ-SLA-01 Priority and due date	Each complaint shall receive Low, Medium, High or Critical priority. A category minimum may raise the indicated priority. Compute and store the due timestamp using priority/category SLA configuration. Later configuration changes shall not alter existing due timestamps.
REQ-SLA-02 Scheduled escalation	A scheduled run shall find past-due SUBMITTED, ASSIGNED, IN_PROGRESS and REOPENED complaints, set escalation, increment level only up to the configured maximum, record time/reason and notify the Department Head. Exclude RESOLVED/CLOSED/REJECTED. Repeat/overlapping runs shall not escalate twice within an interval. Interrupted runs shall resume remaining work without repeating completed escalation. Escalation does not reassign work.
REQ-LIFE-01 Controlled lifecycle	Maintain exactly one state from the state model in SRS §4.3. Allow only its authorized transitions; invalid/unauthorized actions leave state, data and history unchanged. Rejection records a reason. Refuse competing concurrent changes with a modified-record message. Once a complaint leaves SUBMITTED, the Complainant cannot edit/transition it except permitted resolution confirmation or rejection/reopen.
REQ-LIFE-02 History and accountability	Append actor (account, system or anonymous action), previous/new state, timestamp and remarks for every transition, escalation, allocation/reassignment and resolution. No application interface, including administration, shall edit or delete history. Anonymous attribution shall not record identifying data.
REQ-LIFE-03 Resolver resolution	Assigned Department Staff shall progress work and mark IN_PROGRESS complaints RESOLVED only with resolution remarks of at least 20 characters. Optional resolution evidence obeys REQ-CMP-03 and file authorization.
REQ-FDBK-01 Confirm and rate	Only the originating Complainant or valid anonymous reference holder shall confirm/reject a RESOLVED complaint. Confirmation closes it, records actor/time, accepts optional rating 1–5 and notifies the department. This limited anonymous feedback permission does not grant full complaint/evidence access.
REQ-FDBK-02 Reject resolution and reopen	Rejection requires at least 20 characters of reason, returns the complaint to REOPENED in the same department with history retained, increments reopen count and notifies department/Head. Maximum reopens is configurable, default 2; further rejection refuses reopening, informs the Complainant and raises the matter to the Head. Authorized detail/reports retain count and rejection reasons.
REQ-FDBK-03 Automatic closure	A scheduled check shall close complaints remaining RESOLVED without response for a configurable period, default 7 days, and record automatic, unconfirmed closure. It shall not close complaints no longer RESOLVED.

5.3 Assignment and Resolution — notices

ID / capability	Requirement
REQ-NOT-01 Notification events	Create in-app notices for affected parties on submission, department assignment, work commencement, resolution, escalation, reopen and closure. Email affected parties with stored addresses on submission, assignment, resolution, escalation and closure; include complaint ID/status but no credentials, session token or anonymous reference. Anonymous reporters receive no email because no address is stored.
REQ-NOT-02 Email failure handling	Mail failure shall not fail/reverse the complaint action or lose its in-app notice. Retry failed delivery up to 3 times with increasing delay; record permanent failure without changing complaint state.
REQ-NOT-03 Read/unread notices	Display notifications newest first, distinguish unread/read, show unread count in the header and mark viewed notices read.

5.4 Administration and Monitoring

Medium–High priority. Administrator configures users/master data/rules/SLA; Administrator and Head view reports in their authorized scope. F1/F3 configuration, F1 campus reports, F4 dashboards.

ID / capability	Requirement
REQ-ADM-01 Reference data	The Administrator shall create/view/edit/deactivate categories, sub-categories, locations and departments. Active changes are available to new complaints; referenced entries cannot be deleted and existing references are preserved through deactivation.
REQ-ADM-02 Users and membership	Only the Administrator shall manage users, roles and department membership, including assigning/removing Staff/Head membership. Changes apply to permissions on the next request; at least one Administrator must remain. Configuration management remains Administrator-only.
REQ-ADM-03 Routing and SLA configuration	The Administrator shall create/edit/reorder/deactivate category/location routing rules, reject duplicate active rules, and configure resolution duration by priority and optionally category, escalation interval and maximum level. Durations must be positive; invalid values receive a permitted-range message.
REQ-ADM-04 Configuration audit	Record each master-data change with Administrator, changed entity, old/new values and timestamp. Role/membership changes are security-relevant events.
REQ-RPT-01 Reports and export	For a selected date range, provide the Administrator campus-wide reports and the Department Head own-department reports: volume by category/location/department, average and median resolution time, open/overdue/escalated/reopened counts, and authorized reopen information. Refuse other-department figures to Heads. CSV export shall contain the same figures and scope as the displayed report.

6. Other Nonfunctional Requirements

Correctness and security take priority, followed by reliability, usability, maintainability and performance beyond these thresholds. Values below are acceptance targets.

6.1 Performance / 6.2 Safety / 6.4 Quality

ID / category	Requirement
NFR-PERF-01 Web response	At 50 concurrent users on the target deployment, 95% of submission/list/detail/tracking requests shall finish server-side within 2 seconds; queue/action requests within 1.5 seconds.
NFR-PERF-02 Escalation and reports	The escalation run shall process 5,000 open complaints within 60 seconds; a report over 12 months shall finish within 5 seconds.
NFR-PERF-03 Upload and page bounds	A file up to 5 MB shall be accepted and acknowledged within 30 seconds on a 2 Mbps uplink. Lists shall load at most 25 records/request.
NFR-SAFE-01 Complaint preservation	Never silently discard an accepted complaint. Keep it recoverable in a responsible queue until terminal; preserve complaint/history/evidence through restart, deployment and container replacement. No role may delete a complaint; terminate only through CLOSED/REJECTED.
NFR-SAFE-02 Emergency notice and run records	Submission screens shall state that SCMS is not an emergency channel and direct users to appropriate emergency contacts for fire/electrical/medical/structural danger. Record escalation-run start/completion/outcome so failures are visible.
NFR-QUAL-01 Correct workflow	All active-rule matches shall route correctly; unmatched complaints reach the default queue. Retries create no duplicates, states remain valid, and every overdue complaint is flagged/escalated within one configured escalation interval.
NFR-QUAL-02 Usability	A first-time Complainant shall register and submit unaided within 4 primary screens. Refusals shall clearly state the problem and corrective action. Support 320 px and larger layouts, text status labels and copyable complaint IDs.
NFR-QUAL-03 Maintainable modules	Keep F1–F4 identifiable and separately unit-testable, with business logic outside templates. Add master data/routing/SLA settings through configuration, not code. Transition/allocation/escalation/configuration records shall have actor/time and no edit/delete interface.
NFR-QUAL-04 Automated validation	Every functional requirement shall have an automated check; every transition rule needs positive and negative checks. Unit tests shall achieve at least 70% line and 60% branch coverage across F1–F4, reported by the coverage tool.
NFR-QUAL-05 Portability and restart	The application shall run without code changes on local SQLite and deployed Docker/PostgreSQL using configuration, and restart without data loss. No availability percentage is committed for the student deployment.

6.3 Security Requirements

6.3.1 Security objectives

Objective	Meaning
SO-1 — Confidentiality/privacy	Protect complaint/evidence access and anonymous-report privacy.
SO-2 — Integrity/accountability	Protect valid workflow, assignment/configuration and permanent history.
SO-3 — Operational continuity	Avoid lost/partial/duplicate complaints and make failed scheduled/email operations visible.

6.3.2 Main security requirements

ID / control	Requirement
REQ-SEC-01 Credential protection	Store passwords only as salted cryptographic hashes. Never store/log/display passwords in recoverable form or log session tokens. Passwords travel only over the protected login/registration channel.
REQ-SEC-02 Authentication and object authorization	Require authentication for protected functions and enforce server-side role, department and complaint ownership checks on every request, independent of UI buttons. Public exceptions are registration/login, anonymous submission/tracking and valid-reference anonymous feedback. Manipulating IDs shall not grant cross-user or cross-department access.
REQ-SEC-03 Input and query safety	Validate/sanitize all user input on the server, escape user text on output and access the database through parameterized Django ORM operations; script or SQL-like input shall not execute or alter query scope.
REQ-SEC-04 Protected communication	Require CSRF tokens on state-changing POST forms; GET shall not mutate complaint/configuration data. Use HTTPS with HTTP redirection and Secure/HttpOnly/SameSite session cookies; transmit email using encrypted SMTP.
REQ-SEC-05 Protected evidence	Verified-size/type files shall not execute. Store outside public web access or serve through a permission check. Only readers authorized for the parent complaint may download; a direct file reference shall not bypass authorization. Anonymous tracking is not full attachment access.
REQ-SEC-06 Privacy and enumeration resistance	Preserve anonymous privacy under REQ-ANON-02. Apply login lockout and anonymous lookup throttling. Account/complaint/reference responses shall reveal no existence beyond the requester's authorized knowledge; unknown and malformed references receive identical no-match.
REQ-SEC-07 Secrets and safe errors	Keep database/mail credentials and signing keys in environment configuration, never version control. Disable deployment debug output and show errors without stack traces, paths, SQL fragments or other internal details.
REQ-SEC-08 Security event records	Record authentication success/failure, lockout, authorization refusals, role and master-data changes with actor/action/target/time as applicable. Log no passwords, session tokens or anonymous reference/identity linkage.

Anonymous feedback is an explicit narrow exception to login, not to validation, CSRF or complaint-specific authorization.

6.5 Business and Domain Rules

ID	Rule / constraint
BR-01	An accepted complaint belongs to exactly one department. Only assigned Department Staff and the authorized Department Head may progress it under §4.3; only its Complainant or valid anonymous reference holder accepts/rejects resolution. A resolver cannot close their own work.
BR-02	Closure is by Complainant/reference-holder confirmation or confirmation-window expiry. Reopening is limited to the configured maximum, default twice; further rejection becomes the Department Head's responsibility.
BR-03	History is permanent and uneditable by all roles. Anonymous complaints cannot be attributed to individuals by any system function/report.
BR-04	Users never set/change their own role; at least one Administrator remains.
BR-05	Referenced master data is deactivated, not deleted. Routing/SLA changes apply only to new complaints. Category minimum priority may raise user indication; derived priority governs the due time.
BR-06	Escalation increases visibility and notifies the Head; it does not automatically reassign work. SCMS records the complaint workflow; actual repair is outside the system.
DOM-01	Categories cover campus facilities such as electrical, plumbing/water, network/IT, housekeeping, furniture, mess/canteen, civil/structural and Other. Departments reflect maintenance trades and are configurable. Exact categories and department values require stakeholder confirmation.
DOM-02	Express location by block, floor and room/area, not geographic coordinates.
DOM-03	SLA durations use hours/days and realistic campus maintenance targets; values require confirmation. Release 1.0 uses elapsed time and does not model working hours/holidays.

7. Other Requirements

Data/configuration obligations are covered above; no additional integration/localization feature is added. Confirm campus master data/SLA values, reporting duration/date boundaries, exact auto-close equality boundary, 5 MB byte convention, due-date policy after reopen, and password-reset assistance before fixed assertions. Do not assume a new due-date reset policy.

Appendix A — Glossary

Term	Meaning
SCMS / SRS / SAD / STP	Complaint system / requirements / architecture-design / test plan.
Complainant	Authenticated originator of a complaint.
Anonymous Complainant	Reporter without account identity, using a reference.
Department Staff	System role for the resolver.
Department Head	Department supervisor; Warden uses this role.
RBAC	Role-Based Access Control.
SLA	Configured elapsed resolution-time target.
Escalation	Overdue flag/level and Head notification, not reassignment.
Tracking reference	One-time issued anonymous lookup/feedback credential; not complaint ID.
Routing rule	Category/location mapping with specificity/precedence.
Status history	Permanent actor/time/action record.
RTM	Requirement → design → implementation → validation map.

Appendix B — Field Layouts

Compact field requirements, not a complete database specification. Mandatory conditional values are stated explicitly; IDs are generated, not entered. File byte convention remains to be confirmed.

B-1 User

Field(s)	Type / limit	Mandatory / constraint
Name / email	Text 100 / 120	Yes / unique email
Password input / stored hash	Input >=8 / hash 128	Input only; salted hash stored, never plaintext
User type / role	Text 20 / 20	Yes; default Complainant; four roles
Hostel block / room	Text 20 / 10	Hostel resident details; otherwise optional
Contact / department / account status / registered on	Digits 15 / Text 50 / Text 15 / Date	Contact optional; department for Staff/Head; status/date generated

B-2 Complaint

Field(s)	Type / limit	Mandatory / constraint
ID / submitter / reference	Generated ID / account ref / 32-character reference	ID yes; account for signed-in intake; reference anonymous only
Category / sub-category / block / floor / room-area	Text 50 / 50 / 20 / 10 / 30	Category/block/room-area yes; others optional
Title / description / priority	Text 120 / 2000 / enum	Yes; description >=20; Low/Medium/High/Critical
Department / routing / assignee	References; default marker	Department yes; applied rule OR default; assignee optional
State / escalation / reopen	State enum / flag+level / count	Generated; reopen default limit 2
Times / resolution / rating	DateTime / Text 2000 / integer	Submit/due yes; resolved/closed conditional; remarks >=20 on resolve; rating optional 1–5

B-3 Attachment

Field(s)	Type / limit	Mandatory / constraint
ID / complaint / generated name	Reference / reference / Text 100	Yes; protected storage
Original name / type / size	Text 255 / Text 30 / bytes	Verified JPEG/PNG/PDF; <=5 MB; <=3 files
Stage / uploader / time	Text 15 / account ref / DateTime	Intake or resolution; anonymous uploader null; time yes

B-4 Feedback and reporting information

Fields	Constraints / report use
Decision / rating / rejection reason / reopen count	Owner/reference feedback; optional 1–5 rating; rejection >=20 chars; count retained.
Date range / group / counts / mean-median / CSV	Category/location/department; open/overdue/escalated/reopened; own-department Head scope; export matches screen. Formulas require confirmation.

Appendix C — Requirement Traceability Matrix

F1–F4 refer to SAD §3. Design numbers are UC-xx (§4.4), also realized by SAD diagrams/contracts. TC suffixes refer to Test Plan TC-01–15. V01–V05 mean VM-01–05. Tests/system checks are Planned / Not Executed; “Pending” means Pending implementation.

Requirement / brief	Arch.	Design UC	Code	Test TC	System / method
REQ-AUTH-01 — Registration	F1	01	Pending	01,02	01,02
REQ-AUTH-02 — Login and lockout	F1	01	Pending	01	01
REQ-AUTH-03 — Session termination	F1	01	Pending	01	01
REQ-ACC-01 — Role and complaint scope	F1	04 , 05 , 06 , 09 , 10	Pending	02,13	02,13
REQ-CMP-01 — Complaint form	F2	02	Pending	03	03
REQ-CMP-02 — Acknowledgement and record	F2	02 , 03	Pending	03,04	03,04
REQ-CMP-03 — Evidence validation	F2	02 , 03 , 06	Pending	05	05
REQ-CMP-04 — No partial or duplicate intake	F2/F3	02 , 03	Pending	03	03 / V01
REQ-ANON-01 — Anonymous intake and reference	F2	03	Pending	04	04
REQ-ANON-02 — Anonymous privacy	F2	03 , 04	Pending	04,13	04,13
REQ-ANON-03 — Restricted anonymous tracking	F2	04	Pending	04,13	04,13
REQ-TRK-01 — Tracking and detail	F4	04	Pending	03,13	03,13
REQ-ROUTE-01 — Automatic routing	F3	05	Pending	06	06
REQ-ASSIGN-01 — Queue and resolver allocation	F3	05	Pending	06,02	06,02
REQ-SLA-01 — Priority and due date	F3	07	Pending	07	07
REQ-SLA-02 — Scheduled escalation	F3	07	Pending	07	07 / V01
REQ-LIFE-01 — Controlled lifecycle	F4	06 , 08	Pending	08,09	08,09 / V01
REQ-LIFE-02 — History and accountability	F4	04–08	Pending	08	08 / V02
REQ-LIFE-03 — Resolver resolution	F4	06	Pending	08,05	08,05
REQ-FDBK-01 — Confirm and rate	F4	08	Pending	09	09
REQ-FDBK-02 — Reject resolution and reopen	F4	08	Pending	09	09
REQ-FDBK-03 — Automatic closure	F4	08	Pending	10	10
REQ-NOT-01 — Notification events	F4	04 , 06 , 07 , 08	Pending	11	11
REQ-NOT-02 — Email failure handling	F4	04 , 06 , 07 , 08	Pending	11	11
REQ-NOT-03 — Read/unread notices	F4	04	Pending	11	11
REQ-ADM-01 — Reference data	F1/F3	09	Pending	12	12
REQ-ADM-02 — Users and membership	F1	09	Pending	12,02	12,02
REQ-ADM-03 — Routing and SLA configuration	F3	09	Pending	12,06,07	12,06,07
REQ-ADM-04 — Configuration audit	F1/F3	09	Pending	12	12 / V02

Appendix C — Requirement Traceability Matrix (continued)

F1–F4 refer to SAD §3. Design numbers are UC-xx (§4.4), also realized by SAD diagrams/contracts. TC suffixes refer to Test Plan TC-01–15. V01–V05 mean VM-01–05. Tests/system checks are Planned / Not Executed; “Pending” means Pending implementation.

Requirement / brief	Arch.	Design UC	Code	Test TC	System / method
REQ-RPT-01 — Reports and export	F1/F4	10	Pending	14	14
REQ-SEC-01 — Credential protection	F1	01	Pending	01	01 / V02
REQ-SEC-02 — Authentication and object authorization	F1–F4	All	Pending	02,13	02,13
REQ-SEC-03 — Input and query safety	F1–F4	All	Pending	03,13	03,13
REQ-SEC-04 — Protected communication	F1–F4	All	Pending	13	13 / V02
REQ-SEC-05 — Protected evidence	F2/F4	02 , 03 , 04 , 06	Pending	05,13	05,13
REQ-SEC-06 — Privacy and enumeration resistance	F1/F2	01 , 03 , 04	Pending	01,04,13	01,04,13
REQ-SEC-07 — Secrets and safe errors	F1–F4	All	Pending	13	13 / V02
REQ-SEC-08 — Security event records	F1–F4	All	Pending	01,02,12	01,02,12 / V02
NFR-PERF-01 — Web response	F1–F4	All	Pending	15	15
NFR-PERF-02 — Escalation and reports	F3/F1/F4	07 , 10	Pending	15	15
NFR-PERF-03 — Upload and page bounds	F2/F4	02 , 04 , 05	Pending	15	15
NFR-SAFE-01 — Complaint preservation	F1–F4	All	Pending	03	03 / V01
NFR-SAFE-02 — Emergency notice and run records	F2/F3	02 , 03 , 07	Pending	03,07	03,07
NFR-QUAL-01 — Correct workflow	F2/F3/F4	02 , 05 , 06 , 07	Pending	03,06,07,08	03,06,07,08 / V01
NFR-QUAL-02 — Usability	F1/F2/F4	01 , 02 , 04	Pending	03	03 / V03
NFR-QUAL-03 — Maintainable modules	F1–F4	All	Pending	08,12	08,12 / V02
NFR-QUAL-04 — Automated validation	F1–F4	All	Pending	01–15	01–15 / V04
NFR-QUAL-05 — Portability and restart	F1–F4	All	Pending	—	/ V01 V04
BR-01 — Responsibility	F1/F3/F4	05 , 06 , 08	Pending	02,06,08,09	02,06,08,09
BR-02 — Closure and reopen	F4	08	Pending	09,10	09,10
BR-03 — Permanent private records	F2/F4	03 , 04 , 06	Pending	04,08,13	04,08,13
BR-04 — Administrative safeguards	F1	01 , 09	Pending	01,12	01,12
BR-05 — Configuration boundaries	F1/F3	05 , 07 , 09	Pending	06,07,12	06,07,12
BR-06 — Escalation and physical work	F3/F4	06 , 07	Pending	07	07 / V02
DOM-01 — Campus configuration	F1/F3	09	Pending	12	12 / V05
DOM-02 — Campus location	F2	02 , 03	Pending	03,04	03,04
DOM-03 — Elapsed-time SLA	F3	07	Pending	07	07 / V05